

Database & Entity Relationship Diagram (ERD)

Project: LearnLens AI

1. Relational Database Schema (`data/learnlens.db`)

```
erDiagram
SESSIONS ||--o{ TRANSCRIPT_SEGMENTS : contains
SESSIONS ||--o{ FRAMES : captures
SESSIONS ||--o{ CONCEPTS : defines
SESSIONS ||--o{ CHAT_MESSAGES : records
SESSIONS ||--o{ QUIZ_QUESTIONS : evaluates
SESSIONS ||--|| LEARNER_PROFILES : tracks
```

```
SESSIONS {
string id PK
string title
string source_platform
string source_url_or_title
real created_at
real updated_at
real duration_sec
integer progress_pct
string status
integer is_pinned
text tags
integer topic_count
integer concept_count
integer screenshot_count
integer question_count
}
```

```
TRANSCRIPT_SEGMENTS {
string id PK
string session_id FK
real timestamp_start
real timestamp_end
string timestamp_formatted
string speaker
text text
real confidence
string topic
string associated_frame_id
}
```

```
FRAMES {
string id PK
string session_id FK
real timestamp_sec
string timestamp_formatted
string image_path
string thumbnail_path
string p_hash
text ocr_text
text visual_description
string category
real importance_score
text concepts
integer is_pinned
}
```

```
}

CONCEPTS {
string id PK
string session_id FK
string name
text definition
text simple_explanation
text deep_explanation
string evidence_timestamp
string evidence_frame_id
text related_concepts
string exam_importance
string difficulty
real confidence
integer is_pinned
}

CHAT_MESSAGES {
string id PK
string session_id FK
string sender
text text
string mode
real timestamp
text evidence
integer is_pinned
}

QUIZ_QUESTIONS {
string id PK
string session_id FK
text question
text options
integer correct_option_index
string concept_tested
string relevant_timestamp
text explanation
integer user_answer_index
integer is_bookmarked
}

LEARNER_PROFILES {
string session_id PK, FK
text concepts_viewed
text concepts_grasped
text concepts_struggling
integer questions_asked_count
integer pinned_count
real practice_accuracy
text weak_areas
text notes_summary
}
```